

Analytical Questions-Corresponding to Questions and Answers in the Report

1. Does the dataset provide evidence of systematic differences in teaching evaluations across professor genders?
2. To what extent does teaching experience—proxied by the number of student evaluations—affect perceived teaching quality?
3. What is the statistical relationship between a professor's average rating and the reported difficulty of their courses?
4. Do instructors with substantial online teaching activity receive ratings that differ meaningfully from those who primarily teach in-person?
5. How strongly is a professor's average rating associated with the proportion of students who report they would take the course again?
6. Are instructors labeled as “hot” (receiving a pepper) systematically rated higher than those who are not?
7. To what degree can average course difficulty alone predict overall teaching ratings within this dataset?
8. How well can a multivariate regression model—incorporating all available features—predict average teaching ratings, and how does it compare to the single-feature model?
9. How accurately can a classification model predict whether an instructor receives a pepper using only their average rating?
10. How does a full-feature classification model perform in predicting pepper status compared to the rating-only model?
11. Do average teaching ratings differ significantly across U.S. states represented in the dataset?